Exercise 3: Worked solutions

Mathematics for Engineers I · Module 62765 · Exercise Sheet 3

From Fundamentals to AI Apocalypse · personal study guide

Contents

Exercise 3.1: Mathematical induction 2

Exercise 3.2: Maps 3

Exercise 3.3: Properties of the binomial coefficient 4

Exercise 3.4: Elementary functions 4

Self-Study 3.5: Mathematical induction 5

Self-Study 3.6: Power sums 5

Self-Study 3.7: Maps 6

Self-Study 3.8: Finite decimals and computer arithmetic 7

Self-Study 3.9: Fibonacci numbers in Pascal's triangle 8

Answers 9

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Exercise 3.1: Mathematical induction

Induction separates an infinite claim into a base case and a step showing each case implies the next.

Connected topic: Mathematical induction (§2.5) gives the two-part proof structure used in both claims.

1a: Prove the square-sum inequality

Try first: write the base case, then add the new final term $(n+1)^2$ to the induction hypothesis.

Answers: see "Exercise 3.1: Mathematical induction" under Answers at the end.

1b: Prove 4 divides $5^n - 1$

Try first: check $n = 1$ and rewrite the next power using $5 \cdot 5^n$.

Answers: see "Exercise 3.1: Mathematical induction" under Answers at the end.

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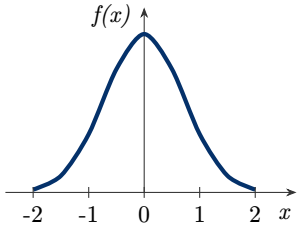
Exercise 3.2: Maps

Injectivity asks whether different inputs can share an output; surjectivity asks whether every target value is reached. A bijection has both properties.

Connected topics: Types of mappings (§3.2), inverse functions (§3.3), and composition (§3.4).

2a: Classify $f(x) = e^{-x^2}$ and $g(x,y) = (4y, x-1)$

Try first: find the range and look for two different inputs with the same output; for g, solve its two coordinate equations.



$f(x) = e^{-x^2}$. The even graph shows why $f(x) = f(-x)$; its values stay positive and at most 1.

Answers: see "Exercise 3.2: Maps" under Answers at the end.

2b(i): Decide whether $f \circ g$ can be formed

Try first: compare the output type of the inner map g with the input type required by the outer map f.

Answers: see "Exercise 3.2: Maps" under Answers at the end.

2b(ii): Find $f \circ f$

Try first: substitute $f(x)$ into the outer f and simplify the square in the exponent.

Answers: see "Exercise 3.2: Maps" under Answers at the end.

2b(iii): Find $g \circ g$

Try first: substitute $(4y, x-1)$ for the input pair of the outer g.

Answers: see "Exercise 3.2: Maps" under Answers at the end.

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Exercise 3.3: Properties of the binomial coefficient

All three identities follow from the factorial definition; the last is Pascal's rule.

Connected topic: Binomial coefficients (§2.4) introduces factorials and Pascal's identity.

3i: Prove C(n,n) = C(n,0) = 1

Try first: substitute $k = n$ and $k = 0$ into the factorial definition, using $0! = 1$.

Answers: see "Exercise 3.3: Properties of the binomial coefficient" under Answers at the end.

3ii: Prove C(n,k) = C(n,$n-k$)

Try first: write both sides using factorials and compare the denominator factors.

Answers: see "Exercise 3.3: Properties of the binomial coefficient" under Answers at the end.

3iii: Prove Pascal's identity

Try first: put the two left-hand fractions over the common denominator $(k+1)!(n-k)!$.

Answers: see "Exercise 3.3: Properties of the binomial coefficient" under Answers at the end.

Exercise 3.4: Elementary functions

Substituting x^4 reduces the degree-eight polynomial to a quadratic. The number of real roots of the parameterized quadratic is controlled by its discriminant.

Connected topics: Elementary functions (§3.5) and the quadratic formula and inequalities (§2.3).

4a: Find and factor the real roots of $f(x) = x^8 - 3x^4 - 4$

Try first: set $t = x^4$. Remember t cannot be negative, even though the resulting quadratic has two algebraic roots.

Answers: see "Exercise 3.4: Elementary functions" under Answers at the end.

4b: Choose c so $g_c(x) = 3x^2 + 2x + c$ has exactly one real root

Try first: set the discriminant to zero, then find the repeated root and write the factorization.

Answers: see "Exercise 3.4: Elementary functions" under Answers at the end.

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Self-Study 3.5: Mathematical induction

The finite geometric-sum formula is proved by adding exactly one new power and simplifying over the fixed denominator.

Connected topic: Mathematical induction (§2.5) supplies the base-step method.

5: Prove the finite geometric-sum formula

Try first: use $n = 1$ as the base case; in the step, add x^{n+1} to the assumed sum.

Answers: see "Self-Study 3.5: Mathematical induction" under Answers at the end.

Self-Study 3.6: Power sums

Each identity is proved independently: the induction step adds the new term $(n+1)$, $(n+1)^2$, or $(n+1)^3$ and factors the result.

Connected topic: Induction (§2.5) supports each proof; the sum notation is reviewed in §2.4.

6a: Prove $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

Try first: check $n = 1$, then add $n+1$ to the formula at n.

Answers: see "Self-Study 3.6: Power sums" under Answers at the end.

6b: Prove $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

Try first: after adding $(n+1)^2$, factor out $n+1$ and compare with the target at $n+1$.

Answers: see "Self-Study 3.6: Power sums" under Answers at the end.

6c: Prove $\sum_{k=1}^{n} k^3 = (\frac{n(n+1)}{2})^2$

Try first: add $(n+1)^3$ to the squared induction hypothesis, then factor $(n+1)^2$.

Answers: see "Self-Study 3.6: Power sums" under Answers at the end.

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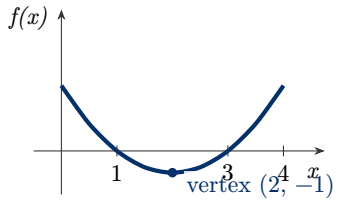
Self-Study 3.7: Maps

Vertex form gives the range and symmetry of the quadratic. For compositions, the inner output must lie in the outer function's domain.

Connected topics: Types of mappings (§3.2), inverse functions (§3.3), and composition (§3.4).

7a: Classify *f*, *g* and *h*

Try first: write f in vertex form, determine the actual range of g, and solve $h(x,y,z)=(u,v)$ for arbitrary u,v.



$\boldsymbol{f(x)=(x-2)^2-1}$**.** The symmetry around $x=2$ explains why f is not injective on all of $\mathbb{R}$; restricting to one side of the vertex makes it invertible.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

7b(i): Form *f* ∘ *s*

Try first: substitute $\sin x$ into f, then find the range by examining the resulting quadratic for $t=\sin x\in[-1,1]$.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

7b(ii): Form *g* ∘ *f*

Try first: require $f(x)\leq 1$ before substituting it under the square root.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

7b(iii): Form *g* ∘ *s*

Try first: check whether every value of $\sin x$ lies in g's domain, then use the extreme sine values for the range.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

7b(iv): Form *g* ∘ *g*

Try first: require the output of the inner g to be at most 1, not just that the original x is at most 1.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

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Self-Study 3.8: Finite decimals and computer arithmetic

Finite decimals form a set closed under addition and multiplication, but they do not include every rational. Truncation gives controlled approximations; floating-point arithmetic can change familiar algebraic laws.

Connected topics: Number systems (§2.1) and real-number arithmetic (§2.2).

8a: Prove D is closed under addition and multiplication

Try first: put two denominators over a common power of 10 for the sum; multiply numerators and denominators for the product.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

8b: Prove $\frac{1}{3} \notin D$

Try first: assume $\frac{1}{3}=\frac{m}{10^n}$ and examine the resulting equality modulo 3.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

8c: Prove the truncation bounds

Try first: apply the defining inequality for $\text{floor}(10^n x)$, then divide by the positive number 10^n.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

8d: Compare the two floating-point sums

Try first: evaluate the parenthesized operations in order and remember that a double cannot represent every integer near 10^{16}.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

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Self-Study 3.9: Fibonacci numbers in Pascal's triangle

The diagonal sums in Pascal's triangle satisfy the same two-step recurrence as the Fibonacci sequence.

Connected topics: Binomial coefficients and Pascal's identity (§2.4), and induction (§2.5).

9a: Compute F_1 through F_{10}

Try first: start from $F_1{=}F_2{=}1$ and add the previous two values at each step.

Answers: see "Self-Study 3.9: Fibonacci numbers in Pascal's triangle" under Answers at the end.

9b: List Pascal's rows 0–7 and compute D_0–D_7

Try first: for each n, add $\mathrm{C}(n{-}k,k)$ across k; use the stated zero convention when $k{>}n{-}k$.

Answers: see "Self-Study 3.9: Fibonacci numbers in Pascal's triangle" under Answers at the end.

9c: Prove $D_n = F_{n+1}$ by induction

Try first: derive $D_{n+2}{=}D_{n+1}{+}D_n$ from Pascal's identity, then use two initial values.

Answers: see "Self-Study 3.9: Fibonacci numbers in Pascal's triangle" under Answers at the end.

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Answers

Exercise 3.1: Mathematical induction

1a · Induction on n

1. **Claim:** $S_n = \sum_{k=1}^{n}(k-1)^2 < \frac{n^3}{3}$.
2. **Base case $n = 1$:** $S_1 = (1-1)^2 = 0 < \frac{1}{3}$, so the claim starts true.
3. **Induction hypothesis:** assume $S_n < \frac{n^3}{3}$ for some $n \in \mathbb{N}$.
4. The next sum adds the term $k = n + 1$, which equals n^2:

$$S_{n+1} = S_n + n^2 < \frac{n^3}{3} + n^2$$

5. Compare the upper bound with the target:

$$\frac{(n+1)^3}{3} = \frac{n^3}{3} + n^2 + n + \frac{1}{3} > \frac{n^3}{3} + n^2$$

6. Therefore $S_{n+1} < \frac{(n+1)^3}{3}$. The base and step prove the claim for every $n \in \mathbb{N}$.

Exercise 3.1: Mathematical induction

1b · Preserve divisibility in the induction step

1. **Base case $n = 1$:** $5^1 - 1 = 4$, divisible by 4.
2. **Induction hypothesis:** suppose $5^n - 1 = 4q$ for some integer q.
3. Then the next expression is:

$$5^{n+1} - 1 = 5 \cdot 5^n - 1 = 5(5^n - 1) + 4$$

4. Substitute the hypothesis: $5(4q) + 4 = 4(5q + 1)$, a multiple of 4. Hence 4 divides $5^n - 1$ for every positive integer n.

Exercise 3.2: Maps

2a · Range and one-to-one tests

1. For f, $x^2 \geq 0$, so $-x^2 \leq 0$ and $0 < \mathrm{e}^{-x^2} \leq 1$. The maximum 1 occurs at $x = 0$; the value 0 is approached as $|x|$ grows but is never reached. Thus range$(f) = (0,1]$, so f is not onto $\mathbb{R}$.
2. Also $f(1) = f(-1) = \mathrm{e}^{-1}$ with distinct inputs, so f is not injective. Therefore it is not bijective as stated.
3. Restrict its domain to $[0,\infty)$ and its range/codomain to $(0,1]$. On this domain f decreases strictly from 1 toward 0, so the restricted map is bijective. Solve $y = \mathrm{e}^{-x^2}$ with $x \geq 0$:

$$\ln y = -x^2 \Rightarrow x = \sqrt{-\ln y}, \text{ for } 0 < y \leq 1$$

   Its inverse is $f^{-1}(y) = \sqrt{-\ln y}$.
4. For g, if $g(x_1,y_1) = g(x_2,y_2)$, coordinate equality gives $4y_1 = 4y_2$ and $x_1 - 1 = x_2 - 1$, hence both input coordinates agree. So g is injective.
5. Given any target $(u,v) \in \mathbb{R}^2$, choose $x = v + 1$ and $y = \frac{u}{4}$. Then $g(x,y) = (u,v)$, so g is surjective as well. It is bijective, with:

$$g^{-1}(u, v) = \left(v + 1, \frac{u}{4}\right)$$

Exercise 3.2: Maps

2b(i) · Match codomain to domain

1. g takes a vector in $\mathbb{R}^2$ and returns a vector in $\mathbb{R}^2$. But f expects a real number, an element of $\mathbb{R}$.
2. The codomain $\mathbb{R}^2$ of g is not contained in the domain $\mathbb{R}$ of f. Thus $f \circ g$ is **not defined** with the stated maps.

Exercise 3.2: Maps

2b(ii) · Compose the real-valued map with itself

1. The output of f lies in $\mathbb{R}$, so it is a valid input for f. The composition has domain $\mathbb{R}$ and mapping rule:

$$(f \circ f)(x) = \mathrm{e}^{-(\mathrm{e}^{-x^2})^2} = \mathrm{e}^{-\mathrm{e}^{-2x^2}}$$

2. The inner value $t = f(x)$ ranges over $(0,1]$. Applying f to such a t gives e^{-t^2}, which ranges from e^{-1} (at $t = 1$) up toward, but never reaches, 1 (as t approaches 0).
3. Therefore range$(f \circ f) = [\mathrm{e}^{-1},1)$.

Exercise 3.2: Maps

2b(iii) · Compose the plane map

1. Both copies of g map $\mathbb{R}^2$ to $\mathbb{R}^2$, so the composition is defined on all of $\mathbb{R}^2$.
2. Apply the rule twice:

$$(g \circ g)(x,\, y) \;=\; g(4y,\, x-1) \;=\; (4(x-1),\, 4y-1)$$

3. The result is an invertible affine map, so its range is $\mathbb{R}^2$. Explicitly, any target (u,v) comes from $x = \dfrac{u}{4} + 1$ and $y = \dfrac{v+1}{4}$.

Exercise 3.3: Properties of the binomial coefficient

3i · The two endpoint choices

1. By definition $\mathrm{C}(n,k) = n!/[k!(n-k)!]$. At $k = n$:

$$\mathrm{C}(n,\, n) \;=\; \frac{n!}{n! \cdot 0!} \;=\; 1$$

2. At $k = 0$:

$$\mathrm{C}(n,\, 0) \;=\; \frac{n!}{0! \cdot n!} \;=\; 1$$

3. Each coefficient counts the unique way to choose either all n objects or none.

Exercise 3.3: Properties of the binomial coefficient

3ii · Symmetry under exchanging chosen and omitted objects

1. For $0 \le k \le n$:

$$\mathrm{C}(n,\, n-k) \;=\; \frac{n!}{(n-k)! \cdot (n-(n-k))!} \;=\; \frac{n!}{(n-k)! \cdot k!}$$

2. Factorial multiplication is commutative, so this denominator equals $k!(n-k)!$. That is $\mathrm{C}(n,k)$, proving the identity.
3. Combinatorially, choosing k objects is equivalent to choosing the $n-k$ objects left behind.

Exercise 3.3: Properties of the binomial coefficient

3iii · Add adjacent binomial coefficients

1. Using the factorial definition, the left side is:

$$\mathrm{C}(n,\,k)\,+\,\mathrm{C}(n,\,k\,+\,1)\,=\,\frac{n!}{k!(n\,-\,k)!}\,+\,\frac{n!}{(k\,+\,1)!(n\,-\,k\,-\,1)!}$$

2. Use the common denominator $(k{+}1)!(n{-}k)!$:

$$=\,\frac{(k\,+\,1)n!\,+\,(n\,-\,k)n!}{(k\,+\,1)!(n\,-\,k)!}\,=\,\frac{(n\,+\,1)!}{(k\,+\,1)!(n\,-\,k)!}$$

3. The final fraction is $\mathrm{C}(n{+}1,k{+}1)$. The stated range $0 \le k \le n{-}1$ ensures all factorials above have non-negative integer arguments.

Exercise 3.4: Elementary functions

4a · Reduce to a quadratic in x^4

1. Let $t = x^4 \ge 0$. Then $f(x) = t^2 - 3t - 4$ factors as $(t{-}4)(t{+}1)$.
2. Setting $f(x) = 0$ gives $t = 4$ or $t = -1$. The second value is impossible because $t = x^4 \ge 0$.
3. Thus $x^4 = 4$. Equivalently $x^2 = 2$, so the real roots are $x = \sqrt{2}$ and $x = -\sqrt{2}$.
4. Factor fully over $\mathbb{R}$:

$$x^8\,-\,3x^4\,-\,4\,=\,(x\,-\,\sqrt{2})(x\,+\,\sqrt{2})(x^2\,+\,2)(x^2\,+\,\sqrt{2}x\,+\,1)(x^2\,-\,\sqrt{2}x\,+\,1)$$

5. The remaining quadratics have no real roots: $x^2{+}2$ is always positive, and each $x^2 \pm \sqrt{2}x + 1$ has discriminant $2{-}4 = -2$. So the two displayed linear factors are all the real linear factors.

Exercise 3.4: Elementary functions

4b · One distinct root means a repeated root

1. For $ax^2{+}bx{+}c$ the discriminant is $b^2{-}4ac$. Here $a = 3$ and $b = 2$, so $\Delta = 4{-}12c$.
2. A real quadratic with non-zero leading coefficient has exactly one distinct real root precisely when $\Delta = 0$. Therefore $4{-}12c = 0$, giving $c = \frac{1}{3}$.
3. The repeated root is $-\frac{b}{2a} = -\frac{2}{6} = -\frac{1}{3}$. The corresponding factorization is:

$$g_{1/3}(x)\,=\,3x^2\,+\,2x\,+\,\frac{1}{3}\,=\,3\left(x\,+\,\frac{1}{3}\right)^2$$

4. **Check:** expanding gives $3x^2{+}2x{+}\frac{1}{3}$, and its only distinct root is $-\frac{1}{3}$ (with multiplicity two).

Self-Study 3.5: Mathematical induction

5 · Induction for every $x \neq 1$

1. Fix any real $x \neq 1$. The claim is $\sum_{k=0}^{n} x^k = \dfrac{1-x^{n+1}}{1-x}$.
2. **Base $n = 1$:** $1+x = \dfrac{1-x^2}{1-x}$, since $1-x^2 = (1-x)(1+x)$.
3. **Induction hypothesis:** suppose the formula holds for n. Add the next term:

$$\sum_{k=0}^{n+1} x^k = \frac{1 - x^{n+1}}{1 - x} + x^{n+1}$$

4. Put the terms over one denominator and simplify:

$$= \frac{1 - x^{n+1} + x^{n+1}(1 - x)}{1 - x} = \frac{1 - x^{n+2}}{1 - x}$$

5. This is the desired formula at $n+1$. Thus it holds for every $n \in \mathbb{N}$ and each fixed $x \in \mathbb{R}\backslash\{1\}$.

Self-Study 3.6: Power sums

6a · Sum of the first n integers

1. **Base:** $1 = 1 \cdot \dfrac{2}{2}$.
2. **Assume** $\sum_{k=1}^{n} k = \dfrac{n(n+1)}{2}$. Then:

$$\sum_{k=1}^{n+1} k = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$$

3. This is the same formula with n replaced by $n+1$, proving it for all positive n.

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Self-Study 3.6: Power sums

6b · Sum of the first *n* squares

1. **Base:** $1^2 = 1 \cdot 2 \cdot \frac{3}{6}$.
2. **Assume** $S_n = \frac{n(n+1)(2n+1)}{6}$. Add the next square:

$$S_{n+1} = \frac{n(n+1)(2n+1)}{6} + (n+1)^2$$

3. Factor $n{+}1$ and combine:

$$= \frac{(n+1)[n(2n+1)+6(n+1)]}{6} = \frac{(n+1)(2n^2+7n+6)}{6}$$

4. Since $2n^2{+}7n{+}6 = (n{+}2)(2n{+}3)$, this equals $\frac{(n+1)(n+2)(2n+3)}{6}$, the formula at $n{+}1$.

Self-Study 3.6: Power sums

6c · Sum of the first *n* cubes

1. **Base:** $1^3 = (1 \cdot \frac{2}{2})^2$.
2. **Assume** $S_n = [\frac{n(n+1)}{2}]^2$. The next sum is:

$$S_{n+1} = \left[\frac{n(n+1)}{2}\right]^2 + (n+1)^3$$

3. Factor $(n{+}1)^2$:

$$= (n+1)^2\left[\frac{n^2}{4} + n + 1\right] = \frac{(n+1)^2(n+2)^2}{4} = \left[\frac{(n+1)(n+2)}{2}\right]^2$$

4. This is the claimed formula for $n{+}1$, completing the induction.

Self-Study 3.7: Maps

7a · Injective, surjective and bijective

1. Complete the square for f:

$$f(x) = x^2 - 4x + 3 = (x-2)^2 - 1$$

   Its minimum is -1 at $x = 2$, so range$(f) = [-1,\infty)$. It is not injective because $f(1)=f(3)=0$, and it is not onto $\mathbb{R}$ because it never takes values below -1.

2. Restrict f to $[2,\infty)$ and set its codomain to $[-1,\infty)$. It is then strictly increasing and onto that range. Solving $y=(x-2)^2-1$ with $x\geq 2$ gives the inverse:

$$f^{-1}(y) = 2 + \sqrt{y+1}, \text{ for } y \geq -1$$

3. For $g(x)=\sqrt{1-x}$, the stated domain $x\leq 1$ makes the radicand non-negative. The function is strictly decreasing, so it is injective. Its range is $[0,\infty)$, not all $\mathbb{R}$, so it is not surjective onto $\mathbb{R}$.

4. Keep the domain $(-\infty,1]$ but change the codomain to $[0,\infty)$. This is a bijection. Solving $y=\sqrt{1-x}$ gives $x=1-y^2$, so $g^{-1}(y)=1-y^2$ for $y\geq 0$.

5. For $h(x,y,z)=(x+y,y+z)$, every $(u,v)\in\mathbb{R}^2$ is reached by choosing $(x,y,z)=(u,0,v)$, hence h is surjective.

6. It is not injective: adding $(-t,t,-t)$ to any input leaves both output coordinates unchanged. More explicitly, all preimages of (u,v) have the form $(u-t,t,v-t)$, $t\in\mathbb{R}$. Thus h is not bijective.

Self-Study 3.7: Maps

7b(i) · Compose f with sine

1. s maps $\mathbb{R}$ to $[-1,1]$, which lies in the domain of f. So $f \circ s$ is defined for every real x:

$$(f \circ s)(x) = \sin^2 x - 4\sin x + 3$$

2. Let $t=\sin x$. Complete the square: $t^2 - 4t + 3 = (t-2)^2 - 1$. Its vertex $t = 2$ lies to the right of $[-1, 1]$, so on this interval the parabola only falls: it decreases as t runs from -1 to 1.

3. At $t=1$ its value is 0; at $t=-1$ its value is 8. Since sine attains every value between -1 and 1, the range is $\mathbf{[0,8]}$.

Self-Study 3.7: Maps

7b(ii) · Restrict inputs to g's domain

1. g accepts inputs at most 1. Since $f(x)=(x-2)^2-1$, the composition domain satisfies:

$$(x-2)^2-1 \leq 1 \Leftrightarrow (x-2)^2 \leq 2 \Leftrightarrow 2-\sqrt{2} \leq x \leq 2+\sqrt{2}$$

2. On that interval:

$$(g \circ f)(x) = \sqrt{1-f(x)} = \sqrt{2-(x-2)^2}$$

3. The radicand ranges from 0 at the two endpoints to 2 at $x=2$. Therefore the range is $[0,\sqrt{2}]$.

Self-Study 3.7: Maps

7b(iii) · Sine always stays in the allowed domain

1. For every real x, $\sin x \leq 1$. Thus $s(x)$ belongs to the domain $(-\infty,1]$ of g and $g \circ s$ has domain $\mathbb{R}$.
2. The rule is:

$$(g \circ s)(x) = \sqrt{1-\sin x}$$

3. Since $-1 \leq \sin x \leq 1$, the radicand ranges from 0 to 2, and both endpoints occur. The range is $[0,\sqrt{2}]$.

Self-Study 3.7: Maps

7b(iv) · Restrict the inner output

1. The first g requires $x \leq 1$. Its output $\sqrt{1-x}$ must also be in g's domain, so $\sqrt{1-x} \leq 1$. Squaring non-negative sides gives $x \geq 0$.
2. The composite domain is therefore $[0,1]$, and its rule is:

$$(g \circ g)(x) = \sqrt{1-\sqrt{1-x}}$$

3. As x increases from 0 to 1, the inner square root decreases from 1 to 0, so the outer expression increases continuously from 0 to 1. The range is $[0,1]$.

Self-Study 3.8: Finite decimals and computer arithmetic

8a · Closure of finite decimals

1. Take $\frac{m}{10^n}$ and $\frac{p}{10^r}$ in D, where $m,p\in\mathbb{Z}$ and $n,r\in\mathbb{N}_0$. Let $N=\max(n,r)$. Then:

$$\frac{m}{10^n}+\frac{p}{10^r}=\frac{m\cdot 10^{N-n}+p\cdot 10^{N-r}}{10^N}$$

2. The numerator is an integer and N is a non-negative integer, so the sum is in D.
3. The product is:

$$\left(\frac{m}{10^n}\right)\left(\frac{p}{10^r}\right)=\frac{mp}{10^{n+r}}$$

4. Again the numerator is an integer and the exponent is non-negative; therefore the product is in D.

Self-Study 3.8: Finite decimals and computer arithmetic

8b · A prime factor missing from powers of 10

1. If $\frac{1}{3}$ were in D, then $\frac{1}{3}=\frac{m}{10^n}$ for some integer m and $n\geq 0$. Cross-multiplying gives $10^n=3m$.
2. The right side is divisible by 3. But 10^n leaves remainder 1 when divided by 3: for $n\geq 1$, expand $10^n=(1+9)^n$; every term after the initial 1 is divisible by 3. For $n=0$ the value is also 1. Contradiction.
3. Thus $\frac{1}{3}$ **is not a finite decimal fraction**.

Self-Study 3.8: Finite decimals and computer arithmetic

8c · Bound the discarded tail

1. Let m be the greatest integer not exceeding $10^n x$. By definition, $m\leq 10^n x<m+1$.
2. Divide through by $10^n>0$:

$$\frac{m}{10^n}\leq x<\frac{m}{10^n}+\frac{1}{10^n}$$

3. Since $x_n=\frac{m}{10^n}$, this says $0\leq x-x_n<10^{-n}$.
4. Also x_n has integer numerator m and denominator 10^n, so $x_n\in D$.

Self-Study 3.8: Finite decimals and computer arithmetic

8d · Rounding breaks associativity

1. For $a{=}10^{16}$, $b{=}{-}10^{16}$, and $c{=}1$, the first grouping cancels the large values exactly before adding 1:

$$(a + b) + c = 0 + 1 = 1$$

2. Near 10^{16}, adjacent double-precision values are 2 apart. The exact value $b{+}c{=}{-}(10^{16}{-}1)$ lies halfway between representable values and rounds to -10^{16} under round-to-nearest, ties-to-even.

3. Therefore:

$$a + (b + c) = 10^{16} + (-10^{16}) = 0$$

4. The results differ, so computer floating-point addition is **not associative** in general. The real-number associative law remains true; rounding after intermediate operations is what changes the computed result.

Self-Study 3.9: Fibonacci numbers in Pascal's triangle

9a · Iterate the recurrence

1. Start with $F_1{=}1$ and $F_2{=}1$.
2. Then $F_3{=}1{+}1{=}2$, $F_4{=}2{+}1{=}3$, $F_5{=}3{+}2{=}5$, and $F_6{=}5{+}3{=}8$.
3. Continue: $F_7{=}13$, $F_8{=}21$, $F_9{=}34$, $F_{10}{=}55$.
4. **Answer:** 1, 1, 2, 3, 5, 8, 13, 21, 34, 55.

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Self-Study 3.9: Fibonacci numbers in Pascal's triangle

9b · Read the diagonal sums

1. The first eight rows of Pascal's triangle are:

Row n	**Coefficients $\mathbf{C(n,k)}$, k=0,...,n**
0	1
1	1, 1
2	1, 2, 1
3	1, 3, 3, 1
4	1, 4, 6, 4, 1
5	1, 5, 10, 10, 5, 1
6	1, 6, 15, 20, 15, 6, 1
7	1, 7, 21, 35, 35, 21, 7, 1

2. Compute the specified sums (terms with $k>n-k$ are zero):

$$D_0 = 1$$

$$D_1 = 1$$

$$D_2 = 1 + 1 = 2$$

$$D_3 = 1 + 2 = 3$$

$$D_4 = 1 + 3 + 1 = 5$$

$$D_5 = 1 + 4 + 3 = 8$$

$$D_6 = 1 + 5 + 6 + 1 = 13$$

$$D_7 = 1 + 6 + 10 + 4 = 21$$

3. These are the Fibonacci numbers F_1 through F_8: D_n appears to equal F_{n+1}.

Self-Study 3.9: Fibonacci numbers in Pascal's triangle

9c · Match the Fibonacci recurrence

1. **Base cases:** D_0=C(0,0)=1=F_1, and D_1=C(1,0)+C(0,1)=1+0=1=F_2. Two are needed because both recurrences depend on the previous two terms.
2. For the recurrence, use Pascal's identity inside the sum:

$$D_{n+2} = \sum_{k=0}^{n+1} \mathrm{C}(n+2-k,\, k) = \sum_{k=0}^{n+1} [\mathrm{C}(n+1-k,\, k) + \mathrm{C}(n+1-k,\, k-1)]$$

3. The first sum is D_{n+1} (out-of-range end terms are zero). In the second, set j=k−1; its non-zero terms become C(n−j,j), summing to D_n. Hence D_{n+2}=D_{n+1}+D_n.
4. **Induction step:** assume D_n=F_{n+1} and D_{n+1}=F_{n+2}. Then D_{n+2}=F_{n+2}+F_{n+1}=F_{n+3} by the Fibonacci recurrence.
5. The two base cases and this step prove D_n=F_{n+1} for every n∈$\mathbb{N}_0$.

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